

Independent Audit Checklist

VAM Strategy Backtest Verification

Fill in each check. Mark PASS or FAIL in the P/F column. Attach evidence for any FAIL before signing the final verdict.

Phase 1: Code Review

Step 1 Script (vam_step1_databento.py)

	#	Check	Expected	Actual	P/F
■	1.1	SMA-50 computed on SPY close (rolling 50-day mean)	pd.rolling(50).mean()		
■	1.2	SMA-200 computed on SPY close (rolling 200-day mean)	pd.rolling(200).mean()		
■	1.3	RSI-14 uses Wilder smoothing (ewm alpha=1/14) on SPY	ewm(alpha=1/14)		
■	1.4	Kill switch: VIX ≥ 30 OR SPY < 200-SMA	Both conditions checked		
■	1.5	Defensive requires 2 consecutive days below 50-SMA	Counter resets on above-close		
■	1.6	RSI trim at 75, rebuy at 60 (asymmetric thresholds)	75 sell / 60 buy		
■	1.7	T+1: pending_trade queue, no df.shift() on prices	No look-ahead		
■	1.8	Slippage: 5 bps normal, 20 bps on VIX ≥ 25 or kill	Dynamic per trade		
■	1.9	Commission uses IBKR percentage-based model	$\approx 0.35\%$ of value, min \$1		
■	1.10	Allocation cap at 99% (BULL_100 = 0.99, not 1.0)	0.99 max alloc		

Step 2 Script (vam_step2_databento.py)

	#	Check	Expected	Actual	P/F
■	1.11	7 states present: BULL_100, BULL_TRIMMED, DEFENSIVE_SPY, DEFENSIVE_QQQ, DEFENSIVE_BOTH, SMA_RECOVERY, CASH	All 7 in enum		
■	1.12	SPY and QQQ defensive triggers are independent	Separate state per instrument		
■	1.13	TQQQ split adjustments: 2021-01-21, 2022-01-13, 2025-11-20	3 splits in TQQQ_SPLITS		
■	1.14	BULL_100 allocation = 74.25% UPRO + 24.75% TQQQ = 99%	74.25 + 24.75 = 99%		

	#	Check	Expected	Actual	P/F
■	1.15	75:25 ratio maintained ($74.25/24.75 = 3.00$ exactly)	Ratio = 3.0		

Phase 2: Data Verification

	#	Check	Expected	Actual	P/F
■	2.1	SPY_daily.csv covers 2019-02-15 to 2025-12-30 (no gaps)	~1,728 rows		
■	2.2	UPRO_daily.csv date range matches SPY (no missing dates)	Same row count as SPY overlap		
■	2.3	VIX_daily.csv has daily closes with no forward-fill gaps >3 days	Daily VIX closes		
■	2.4	UPRO pre-split price roughly 3× higher before 2018-05-24	~3× price before split date		
■	2.5	UPRO price roughly 2× higher before 2022-01-13 (in adjusted data)	Check Jan 12 vs Jan 14 2022		
■	2.6	TQQQ price roughly 2× higher before each of 3 split dates	3 split points verified		
■	2.7	No duplicate dates in any CSV (unique index)	Zero duplicates		

Phase 3: Re-run and Compare

Re-run all scripts from scratch. Compare output to claimed results. Tolerance: $\pm 0.5\%$.

	#	Check	Expected	Actual	P/F
■	3.1	Step 1 final portfolio value	\$333,662		
■	3.2	Step 1 CAGR	+19.17%		
■	3.3	Step 1 Sharpe ratio	0.594		
■	3.4	Step 1 max drawdown	−38.45%		
■	3.5	Step 1 total trades	90		
■	3.6	Step 2 final portfolio value	\$408,259		
■	3.7	Step 2 CAGR	+22.72%		
■	3.8	Step 2 Sharpe ratio	0.661		
■	3.9	Step 2 max drawdown	−49.86%		
■	3.10	Step 2 total trades	294 (UPRO: 147, TQQQ: 147)		
■	3.11	Step 4 Combined final value	\$296,800		
■	3.12	Step 4 Combined Sharpe	0.503		

Phase 4: Logic Stress Tests

	#	Check	Notes
■	4.1	Kill switch activation: Find a day VIX crossed above 30. Verify state = CASH on NEXT trading day (not same day).	Check trade_log state_to
■	4.2	Kill switch exit: Find day VIX dropped below 30 AND SPY recovered above both SMAs. Verify re-entry to SMA_RECOVERY on NEXT day.	Check trade_log state_from=CASH
■	4.3	RSI trim: Find a day RSI-14 crossed above 75. Verify allocation reduced to BULL_TRIMMED on NEXT day.	Check trade_log state_to
■	4.4	RSI rebuy: Find a day RSI dropped below 60 after trim. Verify allocation returned to BULL_100 on NEXT day.	Check trade_log
■	4.5	Defensive entry: Verify 2 consecutive days SPY below 50-SMA triggers DEFENSIVE state on day after second close (not day 1).	Check trade_log state_to
■	4.6	No look-ahead: For any trade, verify signal_date < execution_date. Signal must use only data available at end of signal_date.	signal_date < exec_date
■	4.7	Step 2 independence: Find period where SPY defensive but QQQ is not. Verify UPRO=37.5%, TQQQ=25% (DEFENSIVE_SPY state).	DEFENSIVE_SPY in trade_log
■	4.8	Split adjustment: Check UPRO adjusted price on 2022-01-13. Pre-split prices should be approximately 2× post-split.	~2× price ratio
■	4.9	Manual trade sample: Pick 5 random trades from Step 2 trade_log. Verify entry price = next-day open, slippage correct, commission correct.	5 trades manually verified

Phase 5: Final Verdict

	#	Check	Expected	Actual	P/F
■	5.1	All code review checks passed (Phase 1)	15/15 PASS		
■	5.2	All data verification checks passed (Phase 2)	7/7 PASS		
■	5.3	Re-run results match within 0.5% tolerance (Phase 3)	12/12 PASS		
■	5.4	All logic stress tests passed (Phase 4)	9/9 PASS		
■	5.5	No evidence of look-ahead bias	Clean		
■	5.6	No evidence of survivorship bias	Clean		
■	5.7	Commission model reflects real-world IBKR costs	IBKR ≈0.35% model		

Overall Verdict

PASS	All checks passed. Results are independently verified.
CONDITIONAL	Minor discrepancies found (list below). Results are directionally correct.

FAIL

Material issues found (list below). Results cannot be trusted.

Auditor Signature: _____

Date: _____